Contact

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www.linkedin.com/in/roshinkadannapally (LinkedIn)
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Top Skills

Technical Leadership
Software Architecture
Artificial Intelligence (AI)

Languages

Malayalam (Native or Bilingual)
English (Full Professional)
Hindi (Elementary)

Certifications

Mastering React
The Complete Node.js Course
Elements of AI
Generative AI for Everyone
AI For Everyone

Honors-Awards

Outstanding Graduate Award
National Dean's List 2004-2006
Eta Kappa Nu Association

Publications

FM interference mitigation in a finitely correlated environment using a decorrelating time-varying autoregressive model

Modeling of time-varying Instanteous Frequency in a finitely correlated environment

Implementation of Instantaneous Frequency Estimation based on Time-Varying AR Modeling

Non-stationary interference mitigation in continuous phase modulated signals using estimation subtraction

Roshin Kadanna Pally

Principal Software Engineer and Technical Lead at The MathWorks
Natick, Massachusetts, United States

Summary

I’m a Principal Software Engineer and Technical Lead at MathWorks with a strong track record of delivering impactful software products for engineers and scientists. I specialize in simulation and visualization tools within the MATLAB, Simulink, and RoadRunner ecosystems, and bring a product-focused mindset to solving technical challenges and driving innovation.

My work spans:
▪ Leading cross-functional teams to translate user needs into amazing product features.
▪ Designing APIs and interactive visualization tools for autonomous systems and simulation environments.
▪ Driving product vision from concept to delivery, balancing technical feasibility, customer value, and business objectives.
▪ Full-stack development across JavaScript, C++, Python, and MATLAB, focusing on creating elegant, intuitive, and user-friendly software solutions.

I’m passionate about solving complex challenges and building tools that empower engineers and scientists worldwide to innovate.

Feel free to connect if you’d like to network, discuss product development, or share insights into software architecture and AI-driven solutions.

Experience

MathWorks
16 years 2 months

Principal Software Engineer
May 2022 - Present (4 years 2 months)

Technical lead for Scenario & Scene Authoring, Maps, and Visualization APIs for MATLAB/RoadRunner ecosystem, covering Automated Driving Toolbox, RoadRunner, and RoadRunner Scenarios.

Senior Software Engineer

May 2016 - May 2022 (6 years 1 month)

Greater Boston Area

Developed simulation and visualization tools using JavaScript, C++, and MATLAB

■Developed a JavaScript System Object Editor for MATLAB online

■Built canvas interactions for Scenario Designer and integrated Lidar and INS sensors

■Added lanes ground truth specification, visualization, and detection

■Developed Bird's-Eye Scope data pipeline and Scenario Reader module

Visit ☛ https://www.mathworks.com/videos/driving-scenario-designer-1529302116471.html

Contributed to the full-stack development of Logic Analyzer in Simulink

Visit ☛ http://www.mathworks.com/videos/logic-analyzer-overview-121719.html

Software Engineer

December 2010 - May 2016 (5 years 6 months)

Greater Boston Area

Contributed to the development of the new Simulink Scope and a Unified Scopes infrastructure

Visit ☛ http://www.mathworks.com/videos/new-interface-for-scopes-106836.html

■Developed simulation playback controls, style dialog, and a new programmatic interface

■Added support for sample times, enumerated data types, event-based signals, signal units

■Collaborated with UX and visual design to improve the new Simulink Scope GUI

Developed the System Object Editor front end in Java and MATLAB

Visit ☛ https://www.mathworks.com/help/matlab/matlab_prog/insert-system-object-code-using-matlab-editor.html

Signal Processing and Communications User Interfaces Intern

May 2010 - December 2010 (8 months)

■ Migrated legacy graphics APIs to the new MATLAB graphics system APIs
■ Provided basic graphical property editing capabilities in the Simulink Scope
■ Improved launch performance of Simulink Signal and Scope Manager by more than 50%
■ Added internationalization (i18n) and localization (l10n) support for various GUIs
■ Fixed numerous bugs in the infrastructure and improved usability of GUIs

Virginia Polytechnic Institute and State University
Research Assistant
June 2007 - May 2010 (3 years)
Blacksburg, VA

I received a graduate research scholarship from the DSP Research Lab at Virginia Tech directed by Dr. Louis Beex. I worked on various projects in Signal Processing and Communications. The projects involved research, development, and testing of software to implement signal processing algorithms. Additionally, I helped set up the design project for the DSP & Filter Design course, graded the course projects, and taught classes in the Signals and Systems course.

eMerj
Engineering Intern
August 2006 - October 2006 (3 months)

■ Implemented communication between a micro-controller & fingerprint reader evaluation board (RS-232), Serial EEPROM (I2C)
■ Wrote embedded code to perform capture enrolling and capture verification of fingerprint templates; individually documented the project; team work in integration to an existing device

SUNY New Paltz
Teaching Assistant
August 2005 - December 2005 (5 months)
New Paltz, New York

Course: Computer Simulation Lab
■ Administered MATLAB workshops and lab sessions
■ Guided students in MATLAB graphical user interface development

Education

Virginia Tech

M.S., Electrical Engineering · (2007 - 2009)

SUNY New Paltz

B.S., Electrical Engineering · (2004 - 2006)